**Dataset Description**

**Project Title**

**StockSense AI:** Intelligent Demand Forecasting and Inventory Management System

**Dataset Name**

Retail Store Inventory Dataset

**Dataset Source**

Kaggle

**Dataset Description**

The Retail Store Inventory Dataset contains historical data collected from retail store operations. It includes information about product sales, inventory levels, pricing, orders, and stock movement over time. This data helps us understand customer purchasing patterns and how inventory changes based on demand.

For the **StockSense AI** project, this dataset is used to train a machine learning model that predicts future product demand. These predictions help businesses maintain the right inventory levels, avoid stock shortages, reduce excess inventory, and improve overall inventory management.

**Dataset Features**

- Product ID
- Product Category
- Product Name
- Store ID
- Region / Location
- Inventory Level
- Units Sold
- Units Ordered
- Demand Forecast
- Price
- Discount
- Supplier Information
- Order Date
- Delivery Date
- Stock Status

**Purpose of Using the Dataset**

This dataset is used to build and evaluate an intelligent inventory management system. It helps the model to:

- Analyze historical sales and inventory trends.
- Predict future demand for different products.
- Detect low-stock and overstock situations.
- Suggest when and how much inventory should be reordered.
- Support better inventory planning while reducing operational costs.

**Expected Output**

The system analyzes historical sales and inventory data to predict future product demand. Based on these predictions, it provides useful inventory recommendations such as reorder alerts, suggested reorder quantities, low-stock notifications, and overstock warnings. These insights enable retailers to make informed decisions, maintain optimal inventory levels, improve product availability, and reduce unnecessary inventory costs.